

Data Session 13: The Ballots Themselves

84-355 Democracy's Data | Alaska's cast vote record

Jonathan Cervas

August 11, 2026

Everything here runs as given. Your job is to read what comes out and say what it means.

Where did that number come from?

“He won in the final round after transfers.”

Every session so far has used **totals** — how many people voted for X. Totals are what elections publish, because the ballot is secret and a total is what is left once you throw the ballots away.

Alaska does not throw them away.

Since adopting ranked-choice voting, Alaska publishes its **cast vote record**: one row per scanned ballot, showing every ranking on it. No name, no address — but the actual choices, one voter at a time. It is the closest thing to a public record of individual votes that exists in this country, and almost nowhere else has one.

Today we recount an election from the ballots up, and then check our answer against the state's.

The contest is the 2024 race for Alaska's single seat in the U.S. House: **Mary Peltola**, the Democratic incumbent, against **Nick Begich**, with two minor candidates. Voters could rank up to four.

Part 1 — What one row is

```
s <- read.csv("data/sequences.csv", stringsAsFactors = FALSE)
```

```
nrow(s)
```

```
#> [1] 926
```

```
sum(s$ballots)
```

```
#> [1] 340778
```

```
head(s, 6)
```

```
#>   ballots      rank1      rank2      rank3      rank4 rank5
#> 1  121602      Begich, Nick
#> 2   86278 Peltola, Mary S.
#> 3   17294 Peltola, Mary S.      Hafner, Eric
#> 4   12205      Begich, Nick Howe, John Wayne
#> 5   10674 Peltola, Mary S.      Hafner, Eric Howe, John Wayne Begich, Nick
#> 6    10150
```

340,778 ballots, collapsed into 926 distinct ranking sequences. Every different *pattern* of ranking that any Alaskan produced, and how many people produced it.

Read the first two rows. The most common ballot in Alaska was “**Begich, and nobody else**” — 121,602 of them. The second most common was “**Peltola, and nobody else**” — 86,278.

```
ranks_used <- rowSums(s[, grep("^rank", names(s))] != "")
blank <- sum(s$ballots[ranks_used == 0])
one <- sum(s$ballots[ranks_used == 1])
voted <- sum(s$ballots) - blank

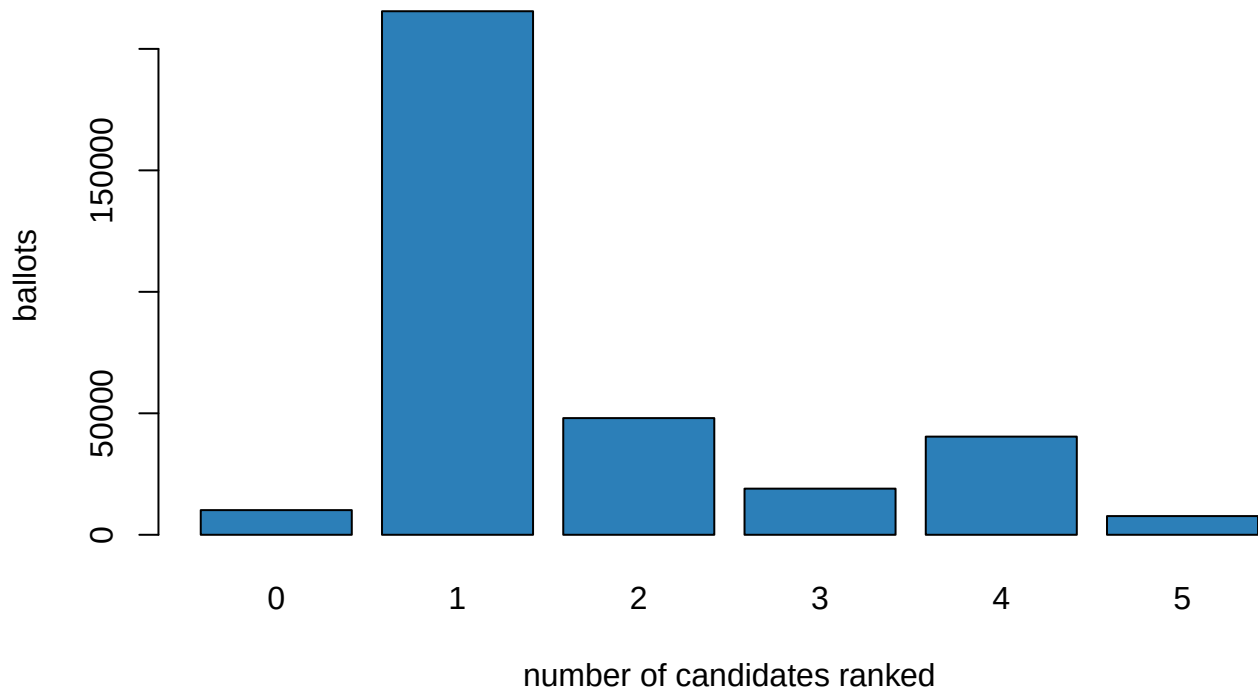
c(blank_on_this_race = blank,
   ranked_only_one = one,
   pct_of_voters = round(100 * one / voted, 1))
```

```
#> blank_on_this_race   ranked_only_one   pct_of_voters
#>           10150.0           215501.0           65.2
```

Given four rankings, 65% of Alaskans used one.

That is the first real finding, and it is a fact about ranked-choice voting that neither its advocates nor its opponents advertise. The system’s whole promise is that you can express a second preference without wasting your vote. Two-thirds of voters declined.

```
u <- tapply(s$ballots, ranks_used, sum)
barplot(u, names.arg = names(u), col = "#2c7fb8",
        xlab = "number of candidates ranked", ylab = "ballots")
```



Part 2 — Who ranked whom

```
top <- aggregate(ballots ~ rank1 + rank2, data = s[s$rank2 != "", ], sum)
top <- top[order(-top$ballots), ]
head(top, 6)
```

```
#>      rank1      rank2 ballots
#> 11 Peltola, Mary S.   Hafner, Eric  39684
#> 13   Begich, Nick Howe, John Wayne  24386
#> 17 Peltola, Mary S. Howe, John Wayne  13890
#> 5  Peltola, Mary S.   Begich, Nick  10771
#> 25   Begich, Nick Peltola, Mary S.   6663
#> 3  Howe, John Wayne   Begich, Nick   4017
```

The most common pair is **Peltola then Hafner** — 39,684 ballots — followed by **Begich then Howe**, 24,386. Voters who bothered with a second choice mostly moved from a major candidate to a minor one.

```
maj <- c("Begich, Nick", "Peltola, Mary S.")
cross <- top[top$rank1 %in% maj & top$rank2 %in% maj, ]
cross
```

```
#>      rank1      rank2 ballots
#> 5  Peltola, Mary S.   Begich, Nick  10771
#> 25   Begich, Nick Peltola, Mary S.   6663
#> 1    Begich, Nick   Begich, Nick   2455
#> 29 Peltola, Mary S. Peltola, Mary S.   1049
```

Two things in that table, and the second one is not what we went looking for.

First, the crossover. 10,771 ballots ranked Peltola then Begich, and 6,663 ranked Begich then Peltola — **17,434 in all, about 5% of the electorate.** Ranked-choice voting is supposed to reward candidates who can be somebody's second choice across the divide. One voter in twenty took the offer.

Second, look at rows three and four. 2,455 ballots rank Begich first and Begich second. Another 1,049 do the same for Peltola. The same person, twice.

```
dup <- top[as.character(top$rank1) == as.character(top$rank2), ]
c(begich = dup$ballots[dup$rank1 == "Begich, Nick"],
  peltola = dup$ballots[dup$rank1 == "Peltola, Mary S."],
  all_candidates = sum(dup$ballots))
```

```
#>      begich      peltola all_candidates
#>      2455      1049      3790
```

3,790 ballots in all name the same candidate twice.

Nothing stopped them: the scanner records the marks it sees, and a voter who wants to make very sure can fill the same column twice. **The tabulator ignores the duplicate** and moves on to the next distinct name — so these ballots count normally — but the behaviour is in the record, and you would never see it in a total. **This is what individual-level data buys you.**

Which brings the argument back to the two-thirds. If ranking one candidate and stopping is *confusion* — people not understanding the offer at the top of the ballot — then the rate should be about the same whichever candidate they marked. Confusion has no reason to prefer a front-runner.

Before you run the next chunk, commit to an answer. It splits that 65.2% by who the voter put first. **Do Begich's voters and Hafner's voters look alike?** Write down two things: whether the gap between them is small — within a few points, which is what confusion would give you — or large; and if it is large, whose supporters stopped after one ranking more often.

```
cands <- c("Begich, Nick", "Peltola, Mary S.", "Howe, John Wayne", "Hafner, Eric")

bul <- data.frame(
  first_choice = sub(",.*", "", cands),
  voters = sapply(cands, function(c_) sum(s$ballots[s$rank1 == c_])),
  pct_bullet = sapply(cands, function(c_)
    round(100 * sum(s$ballots[s$rank1 == c_ & ranks_used == 1]) /
      sum(s$ballots[s$rank1 == c_]), 1)),
  row.names = NULL)

bul <- bul[order(-bul$voters), ]
bul
```

```
#>   first_choice voters pct_bullet
#> 1      Begich 159703      76.1
#> 2      Peltola 152887      56.4
#> 3        Howe  13146      43.4
#> 4      Hafner   3537      35.4
```

They do not look alike at all. 76.1% of Begich's voters ranked nobody else; 35.4% of Hafner's did. The gap is 41 points wide, and it runs the opposite way from the guess most people write

down when they assume the smallest candidate's supporters were the most lost. The pattern runs cleanly down the ballot: the bigger your candidate, the likelier you were to stop after one.

That is not confusion. It is arithmetic voters appear to have done. If your candidate is the front-runner, a second ranking only ever gets read if your first choice is eliminated — and your first choice is not going to be eliminated. If your candidate is polling at one percent, a second ranking is the only part of your ballot that will do anything. There is a second reading this file cannot separate from the first — **campaign instruction**, since “rank one and stop” is a documented tactic in these races — and either way it is something no published total would ever show you.

Part 3 — Count it yourself

Instant-runoff is simple enough to write out. Count everyone's top *continuing* choice; if nobody has a majority, eliminate the lowest and look again at those ballots' next choice; repeat.

```
cols <- grep("^rank", names(s), value = TRUE)

irv <- function(dat, eliminated = character(0)) {
  pick <- rep(NA_character_, nrow(dat))
  for (i in seq_len(nrow(dat))) {
    for (cl in cols) {
      v <- dat[i, cl]
      if (v == "") next
      if (v == "OVERVOTE") break # an overvote stops the ballot
      if (v %in% eliminated) next
      pick[i] <- v; break
    }
  }
  tapply(dat$ballots, pick, sum)
}

elim <- character(0)
for (round in 1:5) {
  cnt <- sort(irv(s, elim), decreasing = TRUE)
  active <- sum(cnt)
  cat(sprintf("\nRound %d (continuing ballots: %s)\n", round,
              format(active, big.mark = ",")))
  print(round(100 * cnt / active, 2))
  if (cnt[1] > active / 2) { cat(" WINNER:", names(cnt)[1], "\n"); break }
  elim <- c(elim, names(cnt)[length(cnt)])
  cat(" eliminate:", names(cnt)[length(cnt)], "\n")
}
```

```
#>
#> Round 1 (continuing ballots: 330,018)
#>      Begich, Nick Peltola, Mary S. Howe, John Wayne      Hafner, Eric
#>      48.39              46.33              3.98              1.07
#>      Write-in
#>      0.23
#>      eliminate: Write-in
#>
#> Round 2 (continuing ballots: 329,614)
```

```

#>      Begich, Nick Peltola, Mary S. Howe, John Wayne      Hafner, Eric
#>      48.49          46.42          4.01          1.08
#> eliminate: Hafner, Eric
#>
#> Round 3 (continuing ballots: 328,300)
#>      Begich, Nick Peltola, Mary S. Howe, John Wayne
#>      48.76          47.01          4.23
#> eliminate: Howe, John Wayne
#>
#> Round 4 (continuing ballots: 322,000)
#>      Begich, Nick Peltola, Mary S.
#>      51.22          48.78
#> WINNER: Begich, Nick

```

Begich wins with 51.22%. You just recounted a congressional election from the ballots.

Part 4 — Check it against the state

```

o <- read.csv("data/official_rounds.csv", stringsAsFactors = FALSE)
o[o$round == "Round 3", ]

```

```

#>      round      candidate votes  pct
#> 8 Round 3      Begich, Nick 164861 51.22
#> 9 Round 3 Peltola, Mary S. 156985 48.78

```

Alaska's official final round: **Begich 51.22%, Peltola 48.78%**. Ours matched to two decimal places.

The **counts** do not match exactly — ours are about 150 ballots higher out of 322,000, or 0.05%. That gap is not a mistake to fix; it is the lab's real lesson. The state's tabulator applies rules ours does not, published on the front page of its own report: *exclude unresolved write-ins, exhausted on two or more ranks skipped*, and a specific treatment of overvotes.

A different, defensible rule about skipped rankings changes the count. It did not change the winner here. In a closer race it could.

Part 5 — Where the losers' votes went

```

tr <- read.csv("data/official_transfers.csv", stringsAsFactors = FALSE)
tr

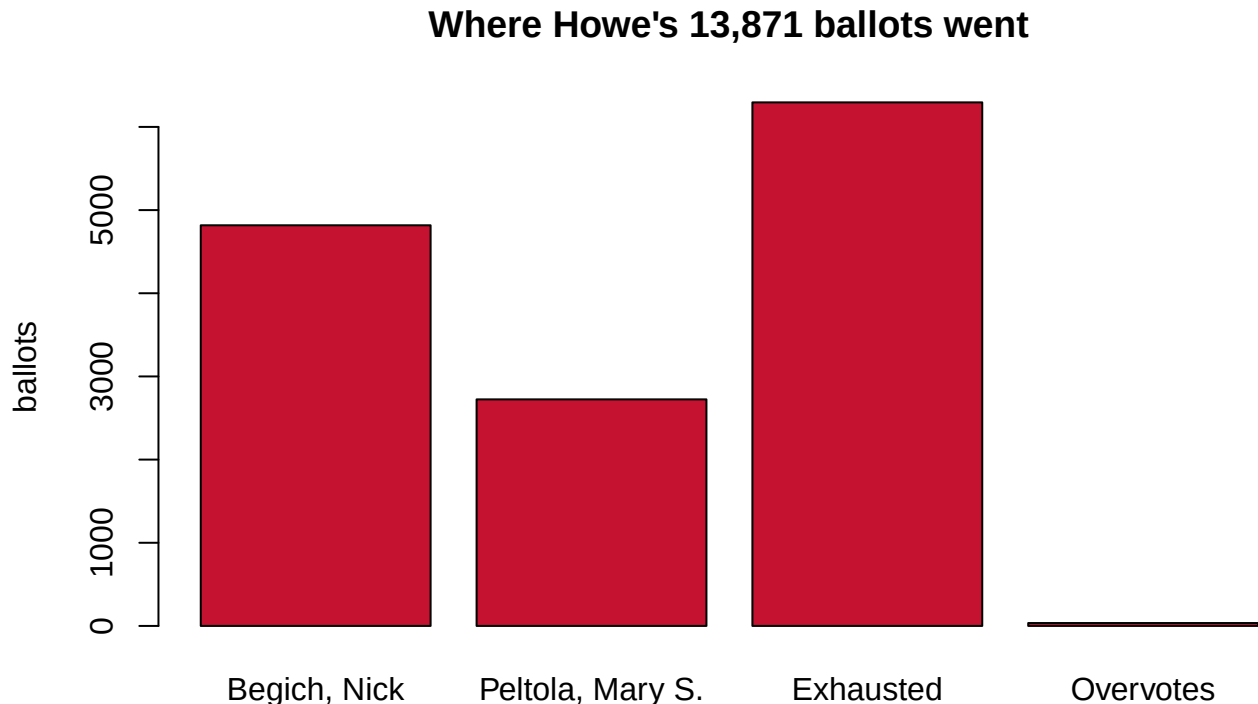
```

```

#>      eliminated transferred_to ballots
#> 1      Hafner, Eric      Begich, Nick      267
#> 2      Hafner, Eric Howe, John Wayne      661
#> 3      Hafner, Eric Peltola, Mary S.      1313
#> 4      Hafner, Eric      Exhausted      1300
#> 5      Hafner, Eric      Overvotes       17
#> 6 Howe, John Wayne      Begich, Nick      4817
#> 7 Howe, John Wayne Peltola, Mary S.      2724
#> 8 Howe, John Wayne      Exhausted      6295
#> 9 Howe, John Wayne      Overvotes       35

```

```
h <- tr[tr$eliminated == "Howe, John Wayne", ]
barplot(h$ballots, names.arg = h$transferred_to, col = "#C41230",
        ylab = "ballots", main = "Where Howe's 13,871 ballots went")
```



When John Wayne Howe was eliminated, his 13,871 ballots split: **4,817 to Begich, 2,724 to Peltola, and 6,295 exhausted.**

Forty-five percent of his voters had ranked nobody else. Their ballots left the count entirely. By the final round **7,595 ballots were exhausted** — people who turned up, voted, and had no say in the decisive round, because the candidate they backed was gone and they had named no alternative.

That is the trade at the heart of ranked-choice voting. It lets you express more than one preference. It does not make you.

Part 6 — Your turn

Change **one** thing, re-run, and write about what changed. Pick one.

Option A — Abolish the runoff. Who wins under plain plurality — just `rank1`? Compare to the RCV winner and say which result better reflects what Alaskans wrote on their ballots. Both answers are defensible; the argument is the work.

Option B — Exhaustion by candidate. For each first-choice candidate, what share of their voters ranked *only* them? (Group `s` by `rank1` and compare `ranks_used == 1` to the rest.) Whose supporters were most likely to bullet vote, and what does that suggest about how each campaign told people to vote?

Option C — A different overvote rule. The `irv()` function stops a ballot at an overvote. Change it to *skip* the overvoted rank and continue to the next one (`if (v == "OVERVOTE") next`). How many ballots does that rescue, and does anything change? Then say which rule you would write into law.

Option D — Your own question. 926 sequences, 340,778 ballots, four candidates.

What this lab cannot tell you

- **Who any of these people are.** No names, no addresses. But a cast vote record is still individual-level: in a small precinct an unusual ranking can narrow down who cast it, which is why the file you have been given is collapsed to sequences and counts, with precinct and tabulator identifiers removed.
- **The certified result.** Alaska says plainly that the CVR **excludes ballots counted by hand** and does not show what write-in voters actually wrote. Our 340,778 is not the official turnout for this contest.
- **Why anyone ranked as they did.** A ballot records a choice and nothing about a reason. Whether 65% bullet voting means satisfaction, confusion, distrust of the system, or a campaign telling people to “rank the red” is not in here — Session 8’s survey data is where that question goes.
- **Whether RCV is good.** One race, one state, one year. What this shows is what the mechanism *does* to a set of ballots, which is a narrower and more useful thing than an opinion about the system.

On the discussion board

By **Mon, Nov 30, 11:59 p.m.** — Thursday is Thanksgiving, so this week’s post runs long. A sentence or two is enough.

Three that would make good posts:

- Given four rankings, two-thirds of Alaskans used one. What do you make of that?
- 6,295 of Howe’s voters had their ballots exhausted before the decisive round. Were they disenfranchised, or did they choose that?
- Our count matched the official percentages exactly and the raw counts by 150 ballots, because of a rule about skipped rankings. Who should decide that rule?

Where the data came from

State of Alaska, 2024 General Election, **cast vote record**, published by the Division of Elections as a Dominion JSON export — 2,043 files, 1.58 GB, one record per scanned ballot. https://www.elections.alaska.gov/results/24GENR/CVR_Export_20241130154411.zip

Official round-by-round results and transfer detail are transcribed from the state’s own **RCV Detailed Report**, <https://www.elections.alaska.gov/results/24GENR/RCV-USRep.pdf>

The committed file contains no individual ballots. 340,778 records were collapsed to 926 distinct ranking sequences with counts, and every precinct and tabulator identifier was dropped, before anything was saved. The build is in `data/build-data.py` — Python rather than R, because base R has no JSON parser and the source is 1.58 GB of JSON.